

Complete Technology Map & Systematic Technical Architecture

A rigorous, end-to-end technical inventory covering confirmed codebase implementations, runtime mechanics, security layers, artificial intelligence modules, testing frameworks, document generators, prototype audits, and our upcoming enterprise production additions.

Document Scope:	Confirmed Stack, Prototype Audit & Evolution Roadmap	Classification:	Official SIH26043 Technical Architecture Spec
Core Runtime:	Node 20 LTS React 19 TypeScript Express.js	Deployment:	Cloudflare Pages + Docker / Render / Vercel Ready
Target Roles:	Students, Industry Mentors, SMEs, College Deans, SuperAdmin	Author / Maintainer:	Skill-Pods Core Engineering & Architectural Guild

Executive Summary & Architectural Scope

This document provides a systematic, line-by-line technical audit of the Skill-Pods platform. It preserves all sections from the official technology map source, categorizes confirmed production technologies versus prototyped/simulated capabilities, formally resolves the truncated Final Architecture blueprint, and details the upcoming enterprise upgrade roadmap (WebSockets, LiveKit SFU, PostgreSQL + pgvector, and live UPI payment gateways).

Systematic Architecture Contents

Sec.	Topic	Key Architectural Scope & Inventory
01	Programming Languages	TypeScript, TSX/JSX, JS, Python, GLSL, HTML, CSS, JSON, TOML, Markdown, XML
02	Frontend Technologies	React 19, Vite, Tailwind CSS 4, Motion v12, Lucide, Native SVG, WebGL HeroShader
03	Backend Technologies	Node.js 20, Express.js REST API, Vite Middleware Mode, TSX Execution, Routing
04	Database & Persistence	Embedded JSON DB (skillpods.json), MongoDB Driver, LocalStorage, In-Memory State
05	Authentication & Security	PBKDF2/SHA-512, Custom JWT (HMAC-SHA256), Google GIS, Rate Limiting, CSP/HSTS
06	AI Technologies	Google Gemini SDK (@google/genai), Guru Copilot, Problem Decomposer, Pitch Deck AI
07	Testing & Validation	Vitest Unit/Integration Tests, TypeScript Strict Compiler, esbuild Production Bundler
08	Build & Deployment	Docker Multi-Stage, Vercel Serverless, Cloudflare Pages, Wrangler, Env Config
09	Python Document Engine	python-docx XML OXML engine, ReportLab PDF generation (SIH presentation & docs)
10	File & Document Tech	HTML specs, PDF artifacts, DOCX proposals, JSON seed data, static asset vault
11	Prototype Audit (Not Fully Implemented)	Real-world audit of README features (WebSockets, WebRTC, Redis, pgvector, UPI)
12	Final Architecture & Data Flow	Complete end-to-end tier data flow (Client -> Gateway -> Security -> Logic -> DB)
13	Confirmed Recent Live Features	Bézier Carousel, 4-Stage Lamp, Direct Messaging, Document Vault, Escrow Simulator
14	Strategic Production Roadmap	5-Phase engineering plan: LiveKit SFU, Postgres+pgvector, Razorpay, GitHub Webhooks

1. Programming Languages

The Skill-Pods codebase employs a cohesive multi-language paradigm designed for strict compile-time type safety, isomorphic JSON serialization between client and server, GPU-accelerated mathematical rendering, and deterministic headless document synthesis.

Language	Primary Scope & Architectural Role	Repository Locations	Status
TypeScript	Primary frontend, backend APIs, shared data types, security token encoding, and build configs.	src/, server/, api/, server.ts, types.ts	CONFIRMED
TSX / JSX	Declarative React UI components, interactive dashboard screens, modals, and SVG vector canvas wrappers.	src/components/, src/App.tsx, src/main.tsx	CONFIRMED
JavaScript (ESM/CJS)	Transpiled browser runtime bundle, esbuild CommonJS backend bundle, and build tool configuration scripts.	dist/server.cjs, dist/assets/*.js	CONFIRMED
Python 3	Headless document generation engine compiling Word (.docx) specifications and vector PDF presentations.	generate_sih_doc.py, generate_sih_pdf.py	CONFIRMED
GLSL (OpenGL ES)	Hardware-accelerated vertex and fragment shaders calculating mathematical color fields and wave dynamics.	src/components/HeroShader.tsx	CONFIRMED
HTML5	Single Page Application (SPA) DOM root shell, Google GIS scripts, meta tags, and static HTML reports.	index.html, public/prototype_dossier.html	CONFIRMED
CSS3	Global design variables, dark cyberpunk lavender palette, glow effects, keyframe animations, glass cards.	src/index.css, component inline styles	CONFIRMED
JSON	Configuration manifests, persistent file database store, mock telemetry payloads, and REST API bodies.	package.json, data/skillpods.json	CONFIRMED
TOML	Cloudflare Pages and Wrangler CLI deployment orchestration and environment bindings.	wrangler.toml	CONFIRMED
Markdown	Project documentation, architectural specifications, SIH pitch details, and memory context files.	README.md, skillpods_gemini.md	CONFIRMED
XML (OXML)	Low-level OpenXML schema nodes modified programmatically for Word table borders, shading, and cell margins.	Inside python-docx automation scripts	CONFIRMED

2. Frontend Technologies

The client architecture is engineered as a responsive, accessible Single Page Application (SPA) leveraging React 19, modern atomic Tailwind styling, and physics-based motion primitives without third-party heavy dependencies.

A. React 19 Core Framework & Lifecycle Architecture

React 19 serves as the foundation for the user interface. It powers all five primary role workspaces (Student, SME Enterprise, Industry Mentor, College Dean, and SuperAdmin). State management avoids cumbersome external state machines (e.g. Redux) in favor of localized, composable React hooks: **useState** for transient view states, **useEffect** for lifecycle data synchronization and API telemetry, **useCallback** for memoized event dispatchers, and **browser localStorage** for persistent cross-session identity. The DOM is mounted via *react-dom/client createRoot* in *src/main.tsx*.

B. Strongly-Typed Domain Modeling (TypeScript 5.x)

Strict static interfaces are enforced throughout the client. Centralized in *src/types.ts*, these interfaces govern data consistency across pods, marketplace capstone projects, cryptographically signed skill passports, mentor milestone gates, college IP registries, and security audit records.

C. Vite 6 Build Engine & Dev Middleware

Vite provides near-instant Hot Module Replacement (HMR) during development and optimizes production output into code-split, treeshaken ES modules. During local development, Vite operates in middleware mode directly within the Express server (*server.ts*), enabling unified full-stack execution on a single port.

D. Tailwind CSS v4 & Cyberpunk Design System

Tailwind CSS v4 is integrated via the official Vite plugin. It enforces consistent layout spacing, responsive flex/grid breakouts, typography hierarchies, and dark glassmorphic cards. The color tokens center around an ultra-dark purple canvas (#08070d, #171422) complemented by vibrant neon lavender glows (#8b5cf6, #6366f1) and accessible slate typography.

E. Motion (Framer Motion v12) Physics Engine

Smooth hardware-accelerated animations are driven by the motion library. Key implementations include: page transitions, modal springs, staggered list entrance animations, telemetry count-ups, and the continuous parabolic Bézier orbital animation showcased in the Innovation Carousel.

F. Native SVG 3D Character Avatars (Architecture Decision)

A deliberate architectural choice was made to implement interactive 3D character avatars (Student, Mentor, SME, College) using layered, mathematically calculated SVG vectors, CSS perspective transforms, and Motion springs rather than Three.js or React Three Fiber. This yields a sub-50KB bundle footprint, zero WebGL context crashes on low-end mobile devices, and instantaneous initial render times.

G. WebGL & GLSL HeroShader Engine

The landing page hero backdrop features a custom WebGL canvas (HeroShader.tsx) running raw GLSL vertex and fragment shaders. It calculates real-time trigonometric wave dynamics and chromatic dispersion directly on the GPU, delivering rich visual aesthetic without CPU main-thread degradation.

H. Native Browser APIs

The frontend relies on standard Web APIs: localStorage for cryptographic token caching; native fetch() for REST API calls; window.location.hash for lightweight, zero-dependency dashboard routing; window.print() for generating official hardcopy skill passports; and Google Identity Services (GIS) client scripts for 1-click Google OAuth.

3. Backend Technologies

The Skill-Pods backend provides a high-throughput, asynchronous RESTful API built on Node.js 20 LTS and Express.js, featuring strict separation of concerns, comprehensive input sanitization, and dual development/production run modes.

A. Node.js 20 LTS & Express.js Application Server

The server application (server/app.ts and server.ts) handles JSON payload parsing, route dispatching, session verification, and static asset streaming. The official containerization targets **Node 20 Alpine** for minimal image size.

B. Comprehensive REST API Endpoints Inventory

HTTP Verb & Endpoint	Controller Handler & Business Purpose	Security Level
POST /api/auth/register	Creates a new user profile, hashes password via PBKDF2, provisions default roles.	Rate Limited
POST /api/auth/login	Validates credentials with timing-safe comparison; issues signed HMAC-SHA256 bearer token.	Rate Limited
POST /api/auth/google	Accepts Google GIS JWT, provisions account or matches existing email, issues session.	Public OAuth
GET /api/auth/me	Validates session bearer token and returns authenticated user identity and claims.	Authenticated
GET /api/marketplace/projects	Retrieves student software projects available for commercial licensing, buyout, or upgrade.	Public / Cached
POST /api/marketplace/inquire	Registers SME commercial buyer licensing inquiries and IP buyout offers.	Authenticated
GET /api/student/skill-passport	Returns verified student passport with SHA-256 integrity hash and mentor signatures.	Authenticated
POST /api/mentor/verify-skill	Authorizes industry mentor verification stamp and attaches verifiable evidence notes.	Mentor Role
GET /api/mentor/milestone-gates	Retrieves active sprint blocking stage gates across assigned student pods.	Mentor Role
POST /api/mentor/milestone-gate/approve	Formally signs off sprint milestone, releasing next development stage & escrow.	Mentor Role
GET /api/college/ip-registry	Returns institutional intellectual property portfolio, commercial valuation, and student inventors.	College Role
GET /api/college/departments	Returns cross-department placement metrics, industry project volume, and faculty citations.	College Role
GET /api/sme/pod-recommendations	Runs AI matching algorithm ranking student pods for SME problem requirements.	SME Role
GET /api/admin/all-chats	SuperAdmin compliance audit endpoint allowing monitored review of 1-on-1 direct messages.	SuperAdmin Only
GET /api/admin/security-audit	Returns immutable event log of authentication attempts, blocked payloads, and IP telemetry.	SuperAdmin Only

C. Dual Runtime Execution Modes (Development vs. Production)

Development Mode: Executed via tsx server.ts. Vite runs as middleware inside Express, dynamically intercepting asset requests and providing instantaneous HMR without prior compilation steps.

Production Mode: The frontend is compiled into dist/ by Vite. The backend is bundled by esbuild into a single standalone CommonJS file (dist/server.cjs) which serves static assets and handles REST APIs with zero development dependencies.

4. Database and Persistence Architecture

Skill-Pods features a flexible dual-persistence strategy designed for frictionless zero-config local evaluation while providing seamless production scale-out via MongoDB Atlas.

Persistence Layer	Implementation Mechanism	Data Types Handled	Status
Embedded JSON DB	Atomic file I/O via Node fs/promises wrapped in server/db.ts; writes to data/skillpods.json.	Users, sessions, audit logs, pods, projects, passports, gates, IP records.	CONFIRMED (Default)

MongoDB & Atlas	Native mongodb driver client; auto-activates when MONGODB_URI is defined in environment.	Document collections mirroring JSON store with BSON ObjectId indexing.	CONFIRMED (Driver)
Browser LocalStorage	Client-side key-value cache in App.tsx and authentication components.	Active JWT token, current user object, UI active tab, local filters.	CONFIRMED
In-Memory State	Module-scoped JavaScript Maps and Arrays in server runtime.	Transient metrics, rate limit sliding windows, live room presence.	CONFIRMED

5. Authentication, Security & Cryptographic Integrity

Security is woven into every layer of the architecture. The platform implements enterprise-grade password hashing, tamper-proof cryptographic token signing, sliding-window rate limiting, and exhaustive input sanitization.

A. Password Hashing: PBKDF2 with SHA-512

Passwords are never stored in plaintext. In server/db.ts, authentication leverages Node's native crypto library utilizing Password-Based Key Derivation Function 2 (PBKDF2) combined with SHA-512, 100,000 iterations, and a 64-byte cryptographically secure pseudorandom salt generated via crypto.randomBytes(64).

B. Custom HMAC-SHA256 Cryptographic Bearer Tokens

Instead of introducing bloated external JWT libraries, the project implements a hardened, zero-dependency token engine in server/security.ts. Tokens are structured as base64url(header).base64url(payload).signature, signed with HMAC-SHA256 using an environment secret (JWT_SECRET). Claims include user ID, role, issue time, and expiration timestamp.

C. Side-Channel Timing Attack Prevention

To defend against timing attacks on password verification and token validation, all buffer comparisons utilize crypto.timingSafeEqual(). This guarantees that comparison execution time remains constant regardless of the number of matching bytes.

D. Google Identity Services (GIS) OAuth 2.0 Integration

Google Sign-In is embedded via the official GIS JavaScript SDK in index.html and handled in LoginPage.tsx. The backend endpoint /api/auth/google receives the credential token, extracts verified claims (email, full name, avatar), provisions the appropriate role, and issues a native platform session token. (Note: In local demo mode, JWT signature verification is handled synchronously; production roadmap introduces official Google Auth Client library verification).

E. Sliding-Window Rate Limiting

To mitigate brute-force credential stuffing and denial-of-service attempts, an in-memory sliding window rate limiter intercepts requests on authentication routes (/api/auth/*). It enforces strict IP-based thresholds (e.g. max 5 failed attempts per 15-minute window) and returns HTTP 429 Too Many Requests upon threshold violation.

F. Enterprise HTTP Security Headers

The security middleware in server/security.ts injects defensive headers on all HTTP responses: **X-Content-Type-Options: nosniff** (MIME-sniffing defense), **X-Frame-Options: SAMEORIGIN** (clickjacking defense), **X-XSS-Protection: 1; mode=block**, **Referrer-Policy: strict-origin-when-cross-origin**, **Permissions-Policy** (camera, microphone restrictions), Content Security Policy (CSP), and production HSTS.

G. Multi-Vector Input Sanitization

All incoming request payloads undergo deep recursive sanitization to neutralize malicious injection: stripping <script> tags and HTML event attributes (XSS defense); stripping javascript: pseudoprotocols; filtering MongoDB query operators prefixed with \$ (NoSQL injection defense); and stripping prototype pollution keys such as __proto__, constructor, and prototype.

H. Immutable Security Audit Logging

All security-critical events (login success, login failure, role changes, blocked injections, CSRF/rate limit triggers) are appended to an immutable audit log (securityLogs collection). Administrators can review incidents in real-time via /api/admin/security-audit.

6. Artificial Intelligence & Intelligent Automation

The platform integrates artificial intelligence to bridge the communication and technical gap between non-technical SME business owners and engineering students.

AI Module	Implementation & Architectural Mechanics	Integration Status
Google Gemini SDK	Integrated via @google/genai package; supports multimodal text and code reasoning using Gemini models.	SDK INSTALLED
GURU AI Co-Pilot	Automated problem decomposer converting vague business pain points into 5 structured developer Kanban tasks.	ACTIVE UI & LOGIC
SME AI Pod Recommendation	Multi-factor heuristic scoring matching open business problems with optimal student pods based on tech stack synergy.	ACTIVE ALGORITHM
Student AI Skill Match	Analyzes student verified competencies and outputs a percentage match against live SME problems with missing skill roadmaps.	ACTIVE ALGORITHM

SME Voice Problem-to-PRD	Speech-to-text pipeline allowing SME founders to speak business pain points and receive structured PRDs with budget estimates.	PROTOTYPE LAB
Investor Pitch Deck AI	Generates 5-slide venture pitch summaries for student capstones to pitch for angel funding or SME buyouts.	PROTOTYPE LAB
AST PR Security Scanner	Simulated static analysis tool inspecting pull request code for OWASP Top 10 vulnerabilities before milestone approval.	PROTOTYPE LAB

7. Testing and Validation Framework

Quality assurance is enforced through automated unit and integration test suites, static TypeScript compiler checks, and production build bundle verification.

A. Vitest Automated Testing Suite

Vitest provides lightning-fast test execution using native Vite transforms. Test suites are located in tests/:

- **api_endpoints.test.ts:** Validates authentication endpoints, registration, login rejection, and role-based permissions.
- **security.test.ts:** Tests PBKDF2 hash uniqueness, HMAC-SHA256 token verification, timing-safe equality, and XSS sanitization.
- **state_persistence.test.ts:** Confirms JSON file database atomic writes, seeded data integrity, and session resumption.

B. TypeScript Strict Static Type Checking

Executed via tsc --noEmit. Configured in tsconfig.json with strict mode enabled, preventing implicit any types, enforcing null safety, and validating API contract signatures.

C. esbuild Production Bundling Verification

The production backend build runs esbuild server.ts --bundle --platform=node --outfile=dist/server.cjs. This compiles TypeScript down to a single optimized CommonJS file, verifying that all imports and server logic bundle without circular dependency or missing module errors.

8. Build and Deployment Technologies

Skill-Pods is engineered for multi-cloud portability, supporting containerized environments, serverless edge networks, and traditional Linux VPS servers.

Deployment Target	Configuration & Automation Files	Execution Mechanism & Architecture
Docker Container	Dockerfile, .dockerignore	Multi-stage build on Node 20 Alpine. Installs packages, builds Vite frontend, bundles backend with esbuild, exposes PORT 3000.
Cloudflare Pages	wrangler.toml, public/_headers, public/_redirects	Edge CDN deployment serving static assets globally; SPA fallback routing via _redirects; security headers via _headers.
Vercel Serverless	vercel.json, api/index.ts	Serverless function adapter wrapping Express API in api/index.ts with automatic rewrites to /api/* and frontend fallback.
Render / Railway	package.json (npm run build & npm start)	Native Node web service running compiled dist/server.cjs connected to MongoDB Atlas via MONGODB_URI.
Environment Config	.env, .env.example	Strict environment variable configuration: PORT, MONGODB_URI, JWT_SECRET, GOOGLE_CLIENT_ID, GEMINI_API_KEY.

9. Python Document Generation Pipeline

Python is utilized in the repository as a specialized headless document compiler producing high-fidelity Microsoft Word dossiers and vector PDF slide decks for Hackathon juries and institutional accreditation audits.

- **python-docx & docx.oxml:** Used in generate_sih_doc.py to build the official Technical Approach document (SkillPods_SIH_Technical_Approach.docx). Uses oxml.parse_xml for granular cell shading and borders.
- **ReportLab:** Used in generate_sih_pdf.py and generate_technology_map_pdf.py to generate publication-quality PDF presentations and technical maps with mathematical coordinate placement, custom palettes, and dynamic two-pass NumberedCanvas page accounting.

10. File and Document Technologies

The repository maintains a structured digital asset vault comprising: **HTML Specifications** (public/prototype_dossier.html, public/sih_presentation_deck.html); **PDF Presentations** (SkillPods_SIH26043_Official_Presentation.pdf); **DOCX Technical Manuals** (SkillPods_SIH_Technical_Approach.docx); **Visual Media** (high-res factory discovery photography, brand icons in public/); **Database Payloads** (mock_database_payloads.json, data/skillpods.json); and **Cloud Infrastructure Manifests** (Dockerfile, wrangler.toml, vercel.json).

11. Prototype Audit: Technologies Mentioned but Not Fully Implemented

In the spirit of absolute transparency and technical rigour required for enterprise reviews and SIH jury defenses, this section audits technologies referenced in marketing or README specifications that currently function as **prototype simulations**, **mock interfaces**, or **planned roadmap milestones** rather than production infrastructure.

Referenced Technology	Current Repository Reality	Codebase Evidence & Analysis	Classification
WebSockets	Simulated via REST polling and in-memory event buffers.	No ws/socket.io server running; telemetry refreshed via interval fetch.	SIMULATED
WebRTC (Video/Audio)	Simulated interactive UI in LivePodRoomModal.tsx.	Uses CSS animated video mockups; no RTCPeerConnection or STUN/TURN server.	SIMULATED
Screen Sharing	Simulated screen-capture toggle.	Does not invoke navigator.mediaDevices.getDisplayMedia().	SIMULATED
Redis	Node.js in-memory Map objects used.	No redis client or connection string configured in backend runtime.	IN-MEMORY
PostgreSQL / TimescaleDB	Not present in database layer.	Data persisted in JSON store / MongoDB; no relational SQL schema active.	ROADMAP
pgvector / Qdrant	Heuristic keyword/tag synergy used.	Vector database embeddings not yet computed via live embedding models.	ROADMAP
MQTT	Not present in current codebase.	No MQTT broker client or IoT telemetry ingestion pipeline installed.	ROADMAP
Razorpay / UPI Gateway	Interactive simulator in EscrowPaymentModal.tsx.	Simulates QR generation and webhook callbacks; no live payment gateway keys.	SIMULATED
GitHub Webhooks / Actions	Simulated PR scanner in GuruCopilotModal.tsx.	No live GitHub App webhook listener or automated CI action runner in repo.	SIMULATED
Cypress / Playwright	Vitest is active; E2E browsers not configured.	No cypress.config.ts or playwright.config.ts present in project root.	ROADMAP
Three.js / React Three Fiber	Replaced by SVG + Motion and WebGL GLSL.	3D character avatars are rendered via native SVG vectors to optimize load time.	ARCH DECISION
Google Maps API	Static campus/enterprise address badges.	No Google Maps JavaScript SDK or Mapbox integration rendered.	NOT IMPLEMENTED
SQLite	JSON flat-file store used instead.	No sqlite3 or better-sqlite3 native binaries compiled in server.	NOT IMPLEMENTED

12. Final Confirmed System Architecture & Data Flow

Source Note: Resolves the truncated heading in original technology-map source documentation.

The confirmed core architecture of Skill-Pods is structured as a resilient, multi-tiered enterprise application spanning four clearly delineated operational strata:

Architecture Tier	Core Technology Stack	Operational Responsibility & Data Flow
1. Presentation & Interaction Tier	React 19, TypeScript, Tailwind CSS 4, Motion v12, Lucide Icons, WebGL GLSL Shaders, SVG 3D Avatars	Renders role-based interfaces (Student, SME, Mentor, College, SuperAdmin). Manages local view state, dispatches asynchronous REST queries, executes orbital Bézier carousels, and renders printable skill passports.
2. Edge Routing & Gateway Tier	Cloudflare Pages Global Anycast CDN, Vercel Serverless Edge, Express Static Middleware	Performs edge TLS termination, asset caching, SPA URL rewriting, and routes API requests to backend endpoints.
3. Security & Application Logic Tier	Node.js 20 LTS, Express.js Router, PBKDF2/SHA-512 Crypto, HMAC-SHA256 Token Engine, Sliding-Window Rate Limiter, Google GIS	Enforces authentication and RBAC roles; sanitizes input payloads against XSS and NoSQL injection; executes milestone gating workflows; calculates pod matching heuristics; maintains security audit logs.
4. Persistence & Intelligence Tier	Embedded Atomic JSON Store (skillpods.json), MongoDB Native Driver / Atlas Cluster, Google Gemini SDK (@google/genai)	Stores authoritative business entities (users, pods, projects, passports, IP registry); executes AI problem decomposition and pitch generation via Gemini API.

13. Confirmed Recent Interactive Features & Upgraded Implementations

Over the recent development sprints, several production-tested interactive features and role utilities were successfully integrated and verified across localhost and Cloudflare Pages deployments:

Interactive Feature Module	Component / Endpoint	Confirmed Implementation & User Experience
Parabolic Bézier Orbital Carousel	src/components/InnovationCarousel.tsx	11 builder cards continuously glide left-to-right along a parametric quadratic Bézier curve with pause-on-hover, soft boundary wrap-around, and apex elevation.
Interactive 4-Stage Wall Torch	src/components/LoginPage.tsx	Realistic overhead wall sconce functioning as a multi-tap lamp (Soft Glow -> Bright -> Turbo High-Beam -> Stealth Darkness OFF) with dynamic light cones.
Cryptographically Stamped Passport	src/components/StudentSkillPassport.tsx	Digital student credential stamped with SHA-256 integrity hash, verified industry mentor seals, and scannable QR verification code.
Academic Scorecard & Document Vault	StudentDashboard.tsx & Lightbox	Official CGPA standing, placement readiness badge, verified semester transcript uploader, and full-screen document lightbox previewer.
Community Network Directory	src/components/CommunityNetworkModal.tsx	LinkedIn-style professional directory allowing Students, SMEs, Mentors, and Admins to browse verified profiles, filter by skills, and inspect portfolios.
Private 1-on-1 Direct Messaging	DirectMessagingModal.tsx & /api/admin/all-chats	End-to-end private messaging between Students, SMEs, and Mentors; SuperAdmin exclusive audit view for compliance and fraud detection.
SME On-Site Discovery Marquee	src/components/HeroImageShowcase.tsx	Continuous animated sliding photo marquee showcasing real-world manufacturing plant discovery visits, sprint lab sessions, and jury defenses.
Escrow & UPI Gateway Simulator	src/components/EscrowPaymentModal.tsx	Simulated milestone funding vault locking bounties in escrow and releasing upon mentor sign-off with simulated UPI QR payment codes.
NAAC & NIRF Accreditation Exporter	src/components/NaacReportModal.tsx	1-click official audit report generation exporting verified industry collaboration metrics mapped to NAAC Criterion 3.5.1 and NIRF parameters.
Cyberpunk 404 Telemetry Radar	src/components/NotFoundPage.tsx	High-tech error view featuring diagnostic telemetry radar, route recovery buttons (Home, Workspace, Community), and fallback handling.

14. Strategic Production Roadmap: The Updated Things We Will Add

To elevate Skill-Pods from an advanced hackathon prototype to an enterprise-grade, nation-scale digital public infrastructure, the engineering team has defined a structured 5-Phase Production Evolution Plan. Each phase systematically addresses and replaces the simulated technologies identified in Section 11.

Phase 1: Real-Time Communication & Telemetry Infrastructure

Goal: Transition from simulated video sprint rooms and interval polling to production real-time infrastructure.

- **Socket.io & WebSocket Gateway:** Deploy a dedicated WebSocket cluster backed by Redis Pub/Sub for instantaneous direct messaging, sprint telemetry, active participant presence, and live Kanban task updates.
- **LiveKit SFU / Coturn STUN/TURN:** Replace simulated video canvas with a production WebRTC Selective Forwarding Unit (SFU) via LiveKit Cloud or self-hosted Mediasoup. Enables sub-100ms multi-party audio/video conferencing, crystal-clear screen sharing, and native audio wave packet analysis.
- **Milestone:** Full production deployment of 4-participant live sprint rooms with zero latency dropouts.

Phase 2: Hybrid Relational & Vector Persistence Engine

Goal: Upgrade persistence from JSON flat-files to an enterprise hybrid database with semantic AI matching.

- **PostgreSQL 16 Multi-Tenant Store:** Establish strongly-typed relational schemas with ACID guarantees for financial ledgers, user authentication, college IP licensing contracts, and milestone gate state machines.
- **pgvector & Qdrant Vector Search:** Compute 768-dimensional text embeddings of student github repositories, coursework transcripts, and SME business problem statements using Gemini Embeddings. Perform cosine similarity searches to deliver sub-second semantic pod recommendations with 95%+ precision.
- **Redis Cache:** Implement Redis cluster for sub-millisecond session caching and sliding-window rate limiting.

Phase 3: Production Payments & Milestone Escrow Smart-Vaults

Goal: Replace simulated UPI payment flows with authorized banking gateway integration.

- **Razorpay / Cashfree Escrow Route:** Implement multi-party split payments where SME problem bounties are held in an RBI-compliant nodal escrow account upon sprint initiation.
- **Automated Milestone Disbursement:** Programmatic fund release triggered automatically when assigned industry mentors digitally sign off on milestone stage gates via `/api/mentor/milestone-gate/approve`.
- **Automated GST & TDS Invoicing:** Instant compilation and dispatch of GST tax invoices, college institutional royalty certificates, and student stipend payout receipts via the Python ReportLab engine.

Phase 4: GitHub App Integration & Automated AST Security CI/CD

Goal: Automate code contribution verification, PR scanning, and milestone gate validation.

- **Official Skill-Pods GitHub App:** Organizations and student pods install the Skill-Pods GitHub App. It listens to pull request webhooks (`pull_request.opened`, `pull_request.synchronize`, `push`).
- **Automated AST & OWASP Security Scanner:** Real AST analysis using Semgrep and SonarQube running in Dockerized worker queues. Automatically audits code for SQL injection, hardcoded secrets, and XSS before mentors review pull requests.
- **End-to-End Test Automation:** Implementation of Cypress and Playwright test suites covering full end-to-end role journeys.

Phase 5: Industrial Hardware & IoT Edge Telemetry Pods

Goal: Expand beyond pure software pods to support manufacturing, logistics, and embedded IoT problem statements.

- **MQTT Telemetry Ingestion:** Implement an EMQX / Eclipse Mosquitto MQTT broker accepting sensor telemetry from SME factory floors (e.g. Bhagyashree Polymers injection molding machines, vibration sensors, energy meters).
- **Edge TimescaleDB Timeseries Store:** Real-time ingestion of high-frequency time-series data for predictive maintenance and yield optimization student capstone challenges.

Institutional Alignment & SIH Jury Defense Conclusion

The Skill-Pods platform embodies the spirit of the National Education Policy (NEP 2020) and the Smart India Hackathon by breaking institutional silos. By uniting verified student engineering talent, SME industrial challenges, and veteran industry mentorship into a mathematically auditable digital ecosystem, Skill-Pods converts classroom academic projects into commercial intellectual property and production-grade software solutions.